REPORT OF THE APPLIED STATISTICS IN DATA SCIENCE PROJECT

COURSE PRESENTER

(DR. ALAA ALMELIBARI)

:DURATION

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INTRODUCTION

Statistical analysis is a process through which the researcher prepares data to conduct analysis on it and extract useful information from it. In our project, we studied the data of the electricity company in the Kingdom of Saudi Arabia during the past four years (2019-2020-2021-2022). Each year contains four regions (central - western - eastern - southern), and each region contains four categories (residential - commercial - industrial - governmental) of electricity users. The mean, variance, standard deviation, correlation, median, mode , quartiles and hypothesis were studied, and based on our study, these laws were applied in the appropriate way to reach accurate results.

ECPLANATION OF CODES

In the first step (using “readxl” to import an excel file and read data from it, it was stored in “read excel” using the “data” variable), and a matrix was created called “categories” containing the types of categories (residential, commercial, government), “industrial” A matrix called “Regions” was created containing a division of the regions to be studied (Central, “Eastern”, “Southern”, “Western”).

In the second step (we started the data analysis phase using a “for loop”, where the “for” goes through the required years and categories, and this loop includes statistics (mean, standard deviation, dispersion, correlation, median value, mode, quartiles, normalization).

In the third step (we began the statistical analysis phase by conducting a t-test for the data and checking for differences in statistical values to make a decision based on statistical evidence).

In the fourth step (we created graphs such as histogram, plot, pie, and barplot to represent and visualize the data for easy understanding and extracting insights from them).

ANALYZING DATA

It appears to us that over the last four years that we studied, the order of electricity consumption rate was as follows:

1. Residential sector: It had the highest electricity consumption rate. This could be due to the Corona crisis and the implementation of the curfew, which caused residents to increase electricity consumption in homes. Another reason is the increased use of electronic devices in the current era and the lack of awareness of rationalization.

2. The industrial sector: comes in second place in electricity consumption. The industrial sector in the Kingdom of Saudi Arabia is witnessing rapid development and prosperity, which leads to an increase in the number of factories and thus an increase in electricity consumption.

3. The commercial sector: The Corona crisis is considered one of the main factors affecting electricity consumption in this sector, as people depend on electronic shopping. Specific times are also set for opening shops, which reduces electricity consumption in them.

4. Government sector: It is considered the sector that consumes the least electricity. This could be due to working hours and the system applied in it, in addition to the nature of workers in this sector.

THE VISION DERIVED FROM THE ANALYSIS

By analyzing this data, we can make predictions about the rate of electricity consumption during the next four years. It is clear that the residential, industrial, and commercial sectors are constantly increasing. As for the government sector, it is in a stable state. It is not clear that it is clearly increasing or decreasing. This increase in electricity consumption depends on many factors. From factors, such as changes in the economy, technology, politics, and environmental factors, some possibilities can be deduced as follows:

Economic Growth: If economic growth continues, demand for electricity in the residential and commercial sectors is likely to increase.

Increased population: As the population increases, there may be an increase in electricity consumption in the residential and commercial sectors due to the increased use of household and commercial appliances.

Industrial sector: The industrial sector may require improvements in energy efficiency to reduce electricity consumption, and this can be achieved by adopting more sustainable and energy-efficient technologies.

Renewable energy and energy efficiency: Governments may adopt policies and legislation that encourage the use of renewable energy sources and enhance energy efficiency, and this may lead to reducing electricity consumption.

Future technology: The development of electrical storage and smart grids may affect the change in the pattern of electricity consumption and enhance energy efficiency.

RECOMMENDATIONS BASED ON THE STUDIED DATA

You can follow these tips to help reduce your electricity consumption:

1. Make it a habit to turn off electrical appliances after you finish using them.

2. Try converting incandescent street lights to LED light bulbs as they consume less energy and last longer.

3. Take advantage of solar energy by installing a solar panel system to generate electricity.

4. Make sure your home is well insulated to reduce energy consumption for heating and cooling.

5. Pay attention to the operating times of air conditioners and use them as sparingly as possible.

6. Set the heater thermostat to a temperature below 55 degrees to save electrical energy.

7. Use energy efficient appliances Look for energy ratings like Energy Star when purchasing new appliances.

COMPARE RESULTS

Code in colab:

Excel file:

THE EXPERIENCES AND SKILLS ACQUIRED BY THE TEAM MEMBERS

Student Name	The works she did	Skills	Challenges
Layan Adnan Al-juhani			
Wed Fawaz Al-shehri	1. I made the presentation 2. I have created a report file 3. Compare the code results by analyzing them in Excel 4. Writing the introduction 5. I created a barcode for the code and the excel file 6. Formatting and writing the report file	(1) Writing the introduction (2) I learned how to write a report	(1)Lack of sufficient experience to handle and analyze data
Noura Nawar Al-huthali	1. I analyzed the results of the code 2. I analyzed the factors that led to an increase in the rate of electricity consumption in each sector 3. Recommendations were made to reduce electricity consumption 4. Forecast of the rate of electricity consumption over the next four years	(1)Lack of time and pressure (2)It is the first time I analyze data	(1) - Experience analyzing data and working on a real analysis project (2) Teamwork
Rahaf Yaseen	1. Description of the data report	(1) Knowledge of writing a descriptive graphic report	Difficulty writing codes that perform the data calculation process
Asayel Sloom Al-harbi	1. Writing the conclusion 2. writing a description of the codes 3. writing a description of the codes	(1) teamwork (2) taking information from the correct source	lack of a strong foundation in R language
Mawadh Saleem Al-oufi	1. Description of the data report	(1)Writing a descriptive report Work with a team	Trying to write a comprehensive report Trying to write codes in R language

CONCLUSION

In this project, we studied, analyzed and visualized electricity consumption data for a group of categories over the years. It became clear in the project that the electricity company plays a decisive role in providing electricity as a primary source of electricity and meets the needs of residents, factories, companies, etc. The electric company intends to face future challenges, including Providing clean and sustainable electricity and increasing efficiency in the production and distribution of electricity. In this project, we tried to clarify the categories that consume the most and the least electricity in order to improve the quality of the electricity company's production by providing it from more than one source, taking into account maintaining a healthy environment and reducing the financial cost.